

INDEX

Sr. No.	Topic	Page No.
1	Abstract	3
2	Introduction	4
3	Problem Statement and Objectives	5
4	Components Used	6 & 7
5	Working Principle	8
6	Circuit Diagram Explanation	9
7	Advantages	10
8	Applications	10
9	Future Scope	10
10	Conclusion	10
11	References	11

ABSTRACT

This project presents the design and implementation of a temperature-controlled fan using a thermistor and a potentiometer. The main objective of this system is to automatically control the operation of a fan based on surrounding temperature conditions.

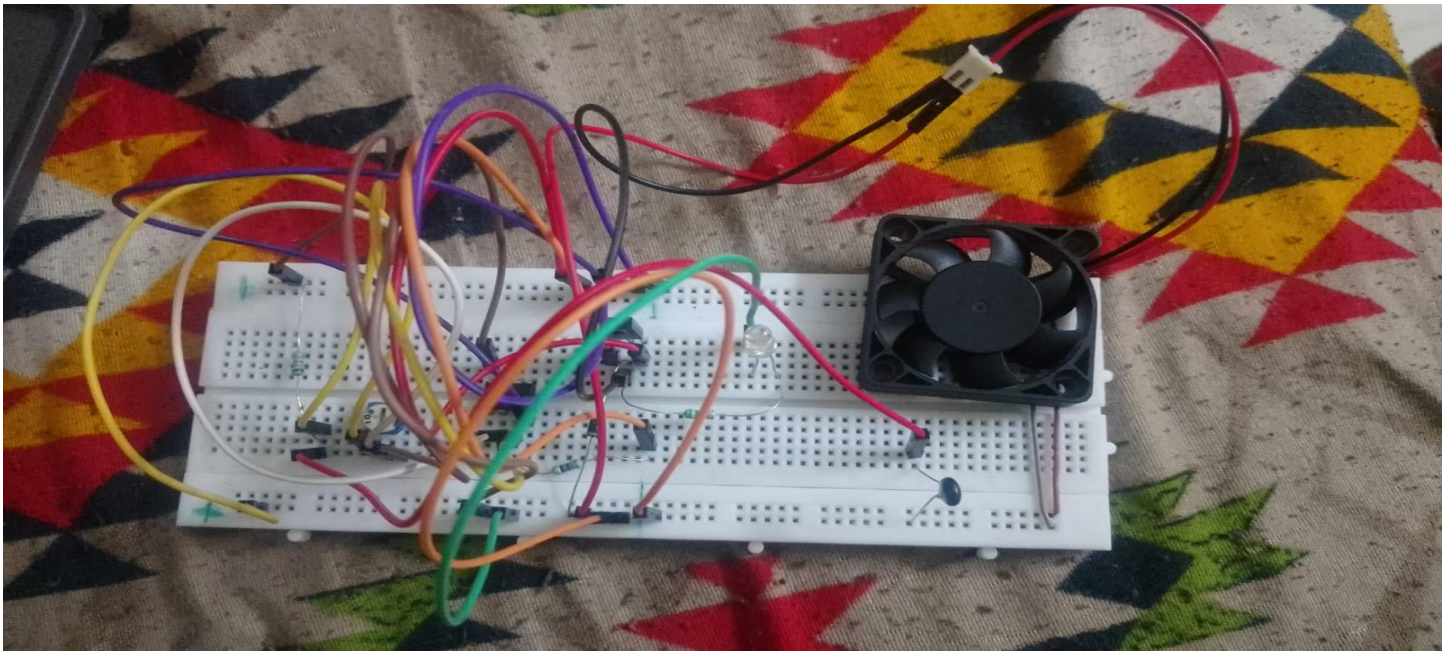
A 10K NTC thermistor is used as a temperature sensor, whose resistance decreases with an increase in temperature. A 10K potentiometer is used to set the reference voltage, allowing adjustment of the temperature threshold. The thermistor and potentiometer together form a voltage divider network.

The output voltage is compared using an LM358 operational amplifier. When the temperature exceeds the set threshold, the comparator activates the fan automatically. When the temperature decreases below the threshold, the fan turns OFF.

This system is simple, cost-effective, and energy-efficient, and it can be used in home appliances, cooling systems, and industrial applications.

Introduction

Temperature regulation is a critical aspect of modern electronic and industrial systems, as excessive heat can significantly affect performance, reliability, and lifespan of devices. Many conventional cooling systems rely on manual operation, which often results in inefficient energy usage and lack of precise control. Therefore, the development of automatic temperature-based control systems has become essential in today's technology-driven environment.



This project presents the design and implementation of a temperature-controlled fan using a 10K NTC thermistor and a 10K potentiometer. The thermistor serves as a temperature sensing element, exhibiting a negative temperature coefficient, wherein its resistance decreases with an increase in temperature. The potentiometer is incorporated to provide an adjustable reference voltage, enabling precise control over the temperature threshold at which the fan operates.

The system utilizes an LM358 operational amplifier configured as a comparator to continuously monitor the voltage variations produced by the thermistor-based voltage divider network. When the sensed temperature exceeds the predefined threshold, the comparator output triggers the fan, ensuring automatic cooling. Conversely, the fan is turned OFF when the temperature falls below the set limit.

This approach offers a simple, cost-effective, and energy-efficient solution for temperature control. The proposed system finds applications in household appliances, electronic cooling systems, and industrial environments where maintaining optimal temperature is crucial for safe and efficient operation.

Problem Statement

In many households and electronic systems, fans are operated manually, which leads to inconvenience and inefficient use of energy. Continuous monitoring of temperature is not always possible, resulting in either overheating or unnecessary power consumption.

Therefore, there is a need to develop an automatic temperature-controlled system that can regulate the operation of a fan based on surrounding temperature conditions without human intervention.

Objectives

- To design and develop an automatic temperature-controlled fan system
- To use a 10K NTC thermistor for sensing temperature changes
- To incorporate a 10K potentiometer for adjusting the temperature threshold
- To implement an LM358 operational amplifier as a comparator
- To reduce energy consumption by eliminating manual control
- To create a low-cost and efficient temperature control system

Components Used

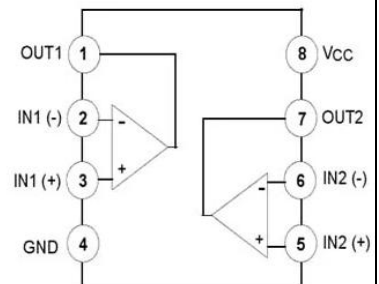
1. **10K NTC Thermistor**: - A 10K NTC Thermistor is a temperature sensor whose resistance is $10k\Omega$ at 25°C and decreases as temperature rises. In a temperature-controlled fan, it is used in a voltage divider circuit. When temperature is low, its high resistance keeps the BD139 transistor OFF, so the fan stays off. As temperature increases, the thermistor resistance drops, raising the control voltage, which turns ON the transistor BD139 and activates the fan or increases its speed. This provides simple automatic temperature-based fan control.



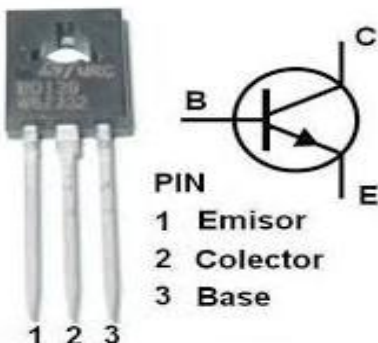
2. **10K Preset Potentiometer**: - A 10K Preset Potentiometer is a small adjustable resistor used to set the temperature threshold in a temperature-controlled fan circuit. It is connected in the voltage divider along with the 10K NTC thermistor. By rotating the preset, the reference voltage can be adjusted, which decides at what temperature the fan should turn ON or change speed. When the voltage from the NTC divider exceeds the set value of the preset, the transistor BD139 activates and switches the fan ON. It allows easy calibration of the circuit without changing fixed resistors.



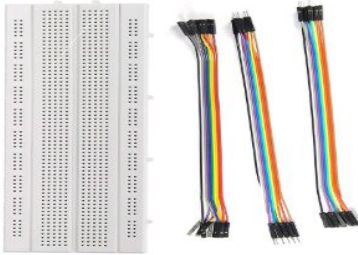
3. **LM358 Operational Amplifier**: - The LM358 is a low-power dual operational amplifier commonly used in temperature-controlled fan circuits for precise and stable switching. It compares the voltage from the 10K NTC thermistor voltage divider with a reference voltage set by the 10K preset potentiometer. When the temperature rises and the NTC voltage exceeds the reference voltage set by the preset, the LM358 output goes high. This output drives a transistor BD39 to turn ON the fan or increase its speed. The LM358 is preferred because it operates on a single supply, has low power consumption, and provides clean switching with minimal components.



4. **BD139 Transistor**: - The BD139 is an NPN power transistor commonly used in temperature-controlled fan circuits as a switch or driver. It is connected to the output of the LM358 op-amp or directly from the NTC voltage divider. When the temperature rises and the control signal (from LM358 or NTC) becomes high, the BD139 turns ON, allowing current to flow through the fan. It can handle higher current (up to 1.5A) and power compared to small signal transistors like BC547, making it suitable for driving 5V DC fans reliably. A base resistor is used to limit the current from the op-amp.

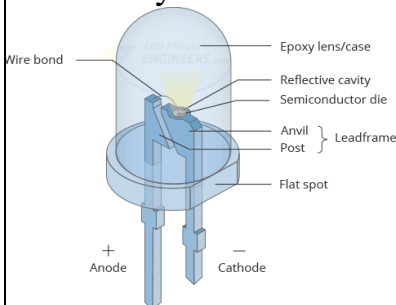
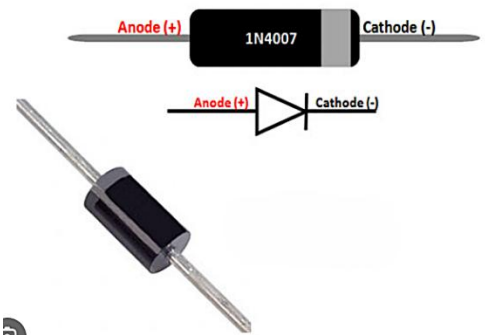


5. **5v DC Fan:** - It acts as the output device and provides cooling when activated.



6. **Breadboard and Connecting Wires:** - Used for assembling and testing the circuit without soldering.

7. **1N4007 Diode:** -The 1N4007 is a general-purpose rectifier diode commonly used in temperature-controlled fan circuits for protection. It is connected across the 5V DC fan terminals in reverse bias (cathode to positive, anode to negative). When the fan is switched OFF, it generates a high voltage spike (back EMF) due to its inductive coil. The 1N4007 diode provides a safe path for this spike to discharge, preventing damage to the transistor (BD139) or op-amp (LM358). It is also known as a flyback or freewheeling diode.



8. **LED (Light Emitting Diode):** -Used as an indicator to show the ON/OFF status of the fan.



9. **9v DC Fan:** - Provides electrical power to the circuit.

10. **Resistors (10K, 4.7K, 560 ohm):** -

- **R1 (10K Ω)**

This resistor forms a voltage divider network with the thermistor. It helps in generating a temperature-dependent voltage which is fed to the comparator.

- **R2 (4.7K Ω)**

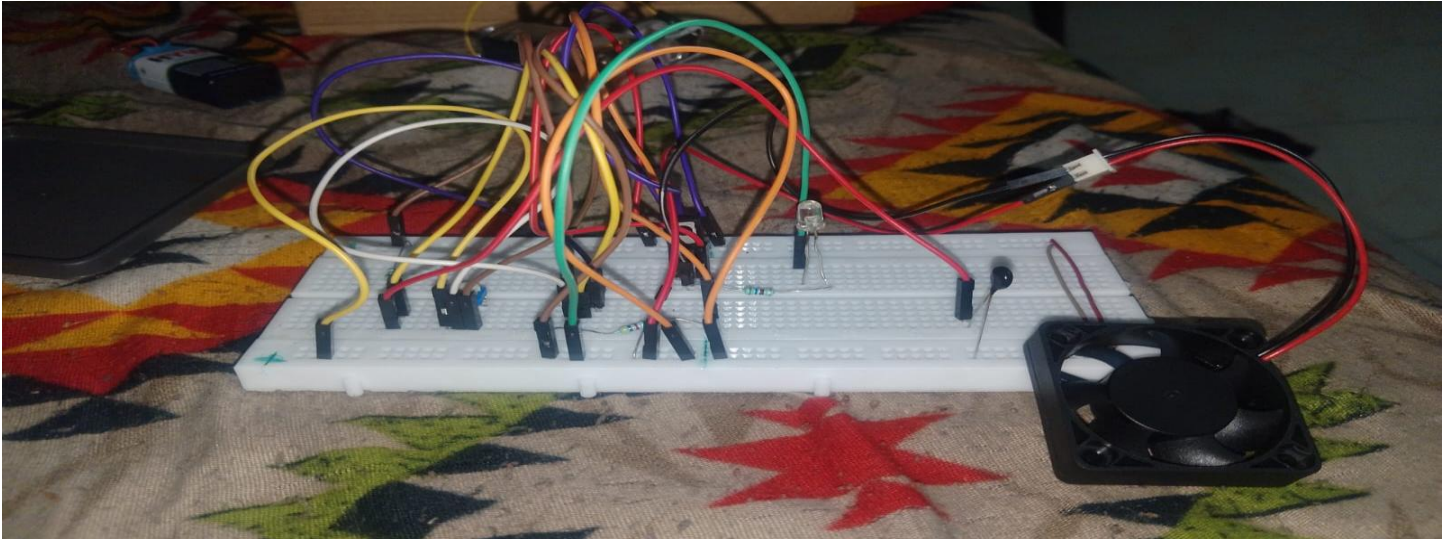
This resistor is connected to the base of the transistor (BD139) and is used to limit the base current, ensuring safe operation of the transistor.

- **R3 (560 Ω)**

This resistor is used with the LED to limit the current and prevent damage to the LED.

Working Principle

The working of the circuit is based on temperature sensing, voltage comparison, and switching action. The **10K NTC thermistor** and **10K resistor (R1)** form a voltage divider network. As the temperature increases, the resistance of the thermistor decreases, which causes a change in the voltage at the input of the LM358 operational amplifier.



A **10K potentiometer** is used to set a reference voltage at the inverting terminal of the LM358. This allows adjustment of the temperature threshold at which the fan should turn ON. The **LM358** is configured as a comparator. It continuously compares the voltage from the thermistor circuit (non-inverting input) with the reference voltage (inverting input).

- When the temperature rises, the voltage from the thermistor side increases/decreases (depending on configuration) and crosses the reference voltage.
- The comparator output then goes HIGH.

This HIGH output is fed through **resistor R2 (4.7K Ω)** to the base of the **BD139 transistor**, which acts as a switch. When the transistor turns ON, it allows current to flow through the **DC fan**, causing it to operate.

An **LED with resistor R3 (560 Ω)** is connected as an indicator to show the ON state of the fan.

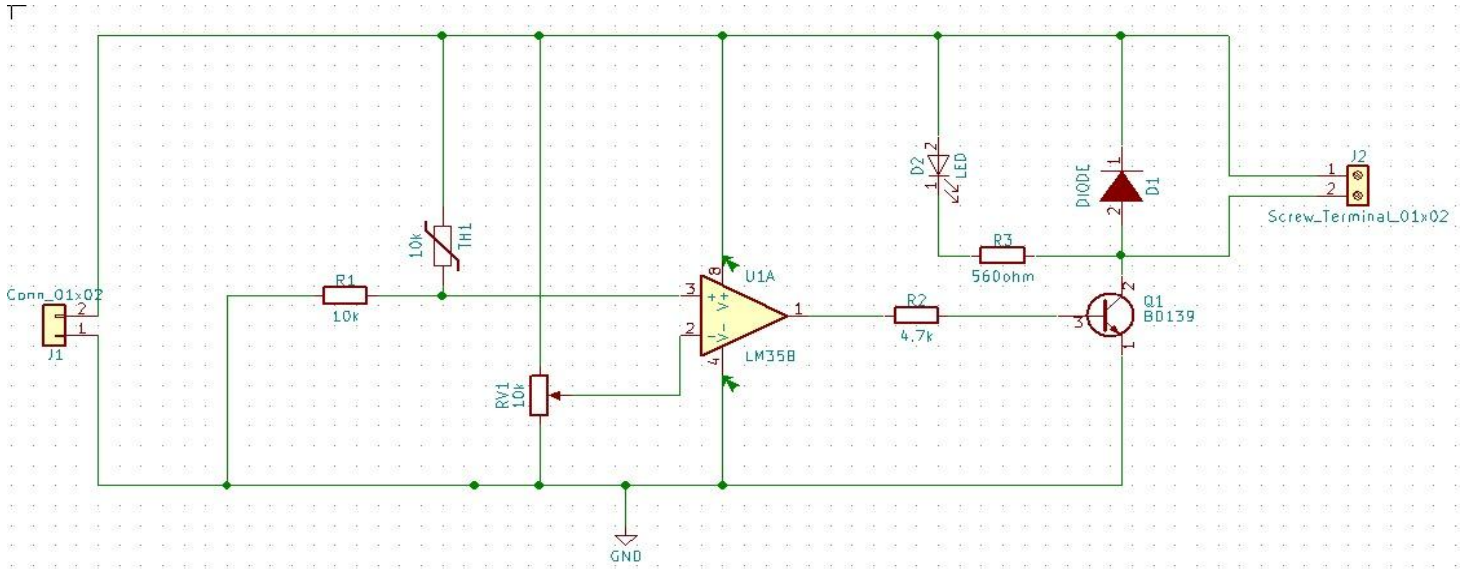
A **1N4007 diode** is connected across the fan to protect the circuit from back EMF generated when the fan is switched OFF.

When the temperature falls below the set threshold, the comparator output goes LOW, the transistor turns OFF, and the fan stops automatically.

Circuit Diagram Explanation

The circuit consists of three main sections: sensing, comparison, and switching.

- **Sensing Section:** - The thermistor (TH1) and resistor (R1) form a voltage divider. This section senses temperature changes and converts them into a voltage signal.
- **Reference Section:** - A 10K potentiometer is used to generate a reference voltage. This allows adjustment of the temperature threshold.
- **Comparison Section:** - The LM358 operational amplifier compares the voltage from the thermistor with the reference voltage.
- **Switching Section:** - The output of the LM358 is connected to the base of the BD139 transistor through resistor R2. The transistor acts as a switch to control the fan.
- **Protection & Indication:** - The 1N4007 diode protects the circuit from back EMF, and the LED with resistor R3 indicates when the fan is ON.



Implementation: - The circuit was implemented on a breadboard using the required components. All connections were made carefully according to the circuit diagram. Initially, the power supply was provided, and the circuit was tested at room temperature. The potentiometer was adjusted to set the desired temperature threshold. To test the system, heat was applied to the thermistor using external sources such as hand heat. As the temperature increased, the fan turned ON automatically. When the temperature decreased, the fan turned OFF. During implementation, proper care was taken to ensure correct polarity of components like the diode, LED, and transistor.

Results / Observations: -

- The circuit successfully detected temperature changes.
- The fan turned ON automatically when the temperature exceeded the set threshold.
- The fan turned OFF when the temperature dropped below the threshold.
- The LED indicator worked properly, showing the ON condition of the fan.
- The system responded quickly and operated reliably.

Advantages

- Automatic operation without human intervention
- Energy efficient system
- Low cost and simple design
- Easy to implement and maintain
- Adjustable temperature threshold using potentiometer

Applications

- Smart home appliances
- Computer and electronic cooling systems
- Industrial temperature control
- Battery cooling systems
- Automatic ventilation systems

Future Scope

- Integration with microcontroller (Arduino) for better accuracy
- Digital temperature display (LCD/7-segment)
- IoT-based remote monitoring and control
- Fan speed control instead of just ON/OFF
- Use of advanced temperature sensors for higher precision

Conclusion

The temperature-controlled fan using a thermistor is a simple and effective solution for automatic temperature regulation. The system successfully detects temperature changes and controls the fan accordingly without manual intervention.

The use of a thermistor, potentiometer, and LM358 comparator makes the circuit cost-effective and easy to implement. The project demonstrates how basic electronic components can be used to design an efficient and practical temperature control system.

Overall, the project achieves its objective of providing an automatic, reliable, and energy-efficient cooling solution.

References

- Boylestad, R.L., *Electronic Devices and Circuit Theory*, Pearson Education.
- Mehta, V.K., & Mehta, R., *Basic Electronics Engineering*.
- Texas Instruments, *LM358 Datasheet*.
- Electronics Tutorials, “Thermistor and Temperature Sensor Basics”.
- Online resources and educational videos related to temperature-controlled fan circuits.